

4 Reductionism

Part 1: Analytic Reductionism and Is-Ought Gap

In the present chapter, I defend two main theses, as against the views of ethical reductionists: that at least some evaluative properties are irreducible, and that moral knowledge cannot be explained without appeal to ethical intuition.

4.1 What is reductionism?

Both ethical naturalism and subjectivism are forms of reductionism about ethics. Reductionism holds that all evaluative properties are reducible to natural properties.

What does that mean? I take 'natural properties' to include all properties (including relational properties) that can be referred to using non-evaluative terms.¹ This includes both physical properties (such as mass, shape, and electric charge) and mental properties (such as believing, feeling sad, and having a tickling sensation). Naturalists and subjectivists disagree with each other over whether certain sorts of mental properties—namely, observers' attitudes towards objects of evaluation—should be included among those to which value properties reduce; but they agree that value properties reduce to some sort of natural properties. To say that an evaluative property 'reduces' to some collection of natural properties is to say that one could explain the nature of the evaluative property—explain what it is—entirely in terms of those natural properties.

Initially, reductionism might seem paradoxical: how could wholly non-evaluative terms be used to explain the nature of value? To see that this is not so paradoxical, consider an analogy: some say that for an object to be red is just for it to reflect predominantly light with wavelengths in the neighborhood of 650 nanometers. Notice that this explanation is given entirely using non-color terms ('reflect', 'wavelength', etc.), so that a person with no previous awareness of color could, if the explanation is correct, come to understand the nature of redness on the basis of it. Whether or not this account of the nature of redness is correct, it is at least not obviously false. Similarly, perhaps it is possible to give an explanation of what goodness is which will not use any evaluative terms.²

A key advantage often claimed for reductionism is that it holds the promise of rendering our knowledge of morality intelligible. If moral properties are irreducible, non-natural properties, then it seems that the only way we might be aware of them is by 'intuition'—a faculty that contemporary philosophers have been reluctant to embrace. But the view that moral properties are reducible to natural properties holds out the hope that we might know moral facts in a manner similar to how we know (other) natural facts, such as by observation or scientific reasoning.³

In the remainder of this chapter, I argue both that reductionism about ethics is false, and that this alleged advantage of the theory is illusory: reductionists cannot escape the appeal to intuition as a source of moral knowledge.⁴

4.2 Analytic reductionism

Analytic reductionism holds that the meaning of any moral term can be given using non-moral terms. For example, an analytic reductionist might hold that 'x is good' is synonymous with 'x increases the total amount of enjoyment in the world'. Or one might hold that 'x is good' is synonymous with 'x is something that we desire to desire'.⁵

G. E. Moore's Open Question Argument refutes analytic reductionism.⁶ Suppose 'N' is an expression containing only non-evaluative terms that denotes some property. 'Are things that have N good?' is an open question. What this means is that a person could coherently give either answer to the question. But 'Are good things good?' is not an open question; a person could not coherently give either answer to that question. Therefore, the two questions are not equivalent. Therefore, 'good things' is not synonymous with 'things that have N'. To illustrate, consider the theory that 'x is good' means 'x increases the amount of enjoyment in the world'. Moore would argue:

1. 'Is it good to do what increases the amount of enjoyment in the world?' is an open question.
2. 'Is it good to do what is good?' is not an open question.
3. Therefore, the questions in (1) and (2) are not synonymous.

4. Therefore, 'increases the amount of enjoyment in the world' is not synonymous with 'is good'.

The idea behind premise (1) is that a person could coherently wonder whether increasing enjoyment is good (after all, an ascetic value system is logically consistent). But as (2) observes, one could not wonder whether doing good is good; no coherent value system denies that doing good is good.

The idea behind (3) is that since one question is open and the other is not, they are not the same question. One might also simply see immediately that the two questions are different, even without invoking the notion of 'open questions'.

Step (4) follows from (3) because the questions in (1) and (2) differ only in that one of them substitutes the predicate 'is good' for the predicate 'increases the amount of enjoyment in the world'; therefore, the difference in meaning between the two questions must be traced to a difference in meaning between those two predicates. If the two predicates had the same meaning, as the reductionist claims, one should be able to substitute one for the other in any sentence without changing the meaning of the sentence.

Moore says that a similar argument will work against any analytic reductionist theory. I know of one apparent exception to this: consider an analytic ideal observer theory, which holds that '\$x\$ is good' means something like, 'An ideally rational, knowledgeable, sensitive being would approve of \$x\$'. The Open Question Argument may fail against this theory, because the first premise is not clearly true.⁷ That is, it is not clear that questions like 'Are the things that an ideally rational, informed, sensitive being would approve of really good?' are open questions. One may think that, provided we characterize ideal observers correctly, an ideal observer is bound to get things right.

I think that there are two ways of understanding this sort of theory, one that makes it non-reductionist and immune to the Open Question Argument, and another that makes it reductionist but vulnerable to the Open Question Argument. On the one hand, if 'ideal rationality' is taken to include the capacity for making rational evaluations, 'ideal knowledge' is taken to include moral knowledge, or 'ideal sensitivity' is taken to include sensitivity to the moral features of things, then I think 'Are the things that an ideally rational, informed, sensitive being would approve of really good?' is not an open question. But in that case, I think we have a non-reductionist theory, since the existence of other evaluative facts or features must be presupposed in defining 'good'. I am not concerned here to refute non-reductionist accounts of value.

If, on the other hand, ideal rationality only includes rationality in the making of non-moral judgments, ideal knowledge only includes non-moral knowledge, and ideal sensitivity only includes non-moral sensitivity, then I think it is an open question whether an ideally rational, knowledgeable, and sensitive being would also possess correct moral beliefs; therefore, it is an open question whether what such a being approved of would be good. Conceivably, a being might be perfectly rational, informed, and sensitive in all non-evaluative areas and yet lack any correct moral judgments. In such a case, this being might fail to approve of what is good, or even approve of things that are bad.

I believe, then, that Moore's argument succeeds against all genuinely reductionist theories of the meanings of evaluative expressions. Reductionists typically grant that if the Open Question Argument succeeds against such theories as the theory that 'good' means 'conducive to pleasure', then it succeeds against all analytic reductionist theories. However, they have said that Moore's argument fails even against the theory that 'good' means 'conducive to pleasure', for at least two reasons.

First, some say that Moore begs the question. Since the reductionist we're confronting holds that the good is just whatever increases enjoyment, obviously he will say that it is not an open question whether it is good to do what increases enjoyment. For Moore to assert otherwise just assumes that the reductionist theory is false.⁸

This response employs an overly expansive conception of 'begging the question'. It is not the case that whenever an argument deploys a premise that directly and obviously contradicts an opponent's position, the argument begs the question. Still less is it true that whenever a consistent opponent would reject at least one of an argument's premises, the argument begs the question. (The latter condition applies to every valid argument.) Consider another famous philosophical argument: Philosopher A claims that 'knowledge' means 'justified, true belief'. Philosopher G points out the following sort of example:

Suppose that Smith justifiably believes that Jones owns a Ford (he has often seen Jones driving a Ford, has seen the title to the car, and so forth). Smith correctly infers from this that the following statement is also true: 'Jones owns a Ford, or Brown is in Barcelona'. (Barcelona was selected randomly; Smith has no idea where Brown is.) But suppose that, improbably enough, Jones actually does not own a Ford; it was sold just a few minutes ago. But by pure coincidence,

Brown happens to be in Barcelona. In this case, Smith believes [Jones owns a Ford, or Brown is in Barcelona]; he is justified in believing this (since he correctly inferred it from a justified belief); and it is true (since Brown is in Barcelona). But intuitively, Smith does not know [Jones owns a Ford or Brown is in Barcelona]. Thus, justified, true belief is not the same as knowledge.

The above argument is widely, and rightly, taken to conclusively refute the 'justified, true belief' account of knowledge; indeed, it is one of the few, and one of the most celebrated, examples of a conclusive refutation of a previously widely-held view in modern philosophy. How weak it would be for \$A\$ to reply:

\$G\$ has merely begged the question. I say that knowledge is justified, true belief. From this, it directly and obviously follows that Smith does know [Jones owns a Ford or Brown is in Barcelona] in \$G\$'s example. For \$G\$ to assert that Smith lacks such knowledge just assumes that my definition is wrong. So \$G\$ has proven nothing.

\$A\$ is correct to note that \$G\$'s premise directly and obviously contradicts \$A\$'s theory. But this does not mean that \$G\$ begged the question; it means only that \$G\$'s refutation of \$A\$ was direct and obvious. \$G\$ succeeds while \$A\$ fails because \$G\$'s premise, once stated, is intuitively obvious, while \$A\$'s theory is not intuitively obvious but needs to be tested by considering examples. \$A\$ thus is not in a position to simply appeal to his theory as a justification for denying the premise that would be used to refute the theory.

Similarly, Moore succeeds against the reductionist because Moore is able to adduce premises that, once stated, are intuitively obvious to competent English speakers, whereas the reductionist's proposed analysis of 'good' is not intuitively obvious. We have a sense of when a statement is contradictory, and 'Enjoyment isn't good' does not sound contradictory. The reductionist thus is not in a position to simply appeal to his theory to justify rejecting the premises Moore would use against it. Nor should the reductionist question the general reliability of our judgments of consistency and inconsistency. For judgments of this kind are the essence of our knowledge of logic, without which none of us are in a position to advance or evaluate any arguments.

The second reply to Moore begins with the observation that even for individuals who understand a given word, it is often difficult offhand to identify the correct definition of that word. For instance, one who knows the word 'circle' may still have difficulty coming up with the definition, 'the set of all points a given distance from a given point'. Thus, a question may often appear to someone to be open even when it is not, simply because the person does not realize what the correct analysis of his own concept is. Perhaps Moore's question, 'Is it good to do what increases the amount of enjoyment in the world?' only appears open to us because we do not at first realize what the correct analysis of 'good' is.[10]

This response rests on an uncharitable reading of Moore. Moore need not assume that the correct analysis of a given concept is always immediately obvious to anyone who has the concept. Instead, Moore can get by with only the assumption that, once a specific definition has been stated, and once competent speakers of the language have thoroughly understood and reflected carefully on that definition, their linguistic intuitions provide at least a *prima facie* guide to the definition's correctness. This assumption is much weaker and more reasonable than the assumption that the analysis of one's concepts is always immediately obvious, and it is difficult to see how anyone engaged in attempting to define words or concepts—as the analytic reductionist is—can reject it. For without it, it is hard to see how we could ever know what the definition of anything is.

The intuitions to which Moore appeals—for example, that 'Pleasure is not good' is consistent, and that 'Is pleasure good?' does not mean 'Is pleasure conducive to pleasure?'[11]—are ones that occur after competent speakers of English have thoroughly understood the definition of 'good' under discussion and have reflected carefully on it. Therefore, these intuitions provide at least a *prima facie* test of the definition's correctness. That is, in the absence of specific, independent grounds for rejecting these intuitions or holding them to be unreliable, they provide adequate reason for rejecting the reductionist's definition. Since reductionists have in fact provided no such grounds, we should reject the proposed definition.[12]

Moore's claim to refute all forms of analytic reductionism rests on his ability to produce similar intuitions for every proposed definition of 'good'. Consider some other potential definitions, and some premises that might be used in Moorean arguments against them:

Proposed Analysis

\$x\$ is good = God approves of \$x\$.

Moorean Premise 'God is good' does not mean 'God approves of himself'.

x is right = x is in accord with the prevailing social customs.

'Is it right to obey social customs?' does not mean 'Does obeying social customs accord with social customs?'

x is good = I approve of x .

'Are the things I approve of really good?' does not mean 'Do I really approve of the things I approve of?'

x is good = x promotes life.¹³

'Is life good?' does not mean 'Does life promote itself?'

I cannot discuss every possible analysis of 'good' here. But the above examples (including the case of the definition involving enjoyment) suggest that Moore could indeed adduce relevant, plausible premises for his argument against any form of analytic reductionism. I am not making an inductive argument based on five cases here. Rather, I am suggesting that upon considering these five cases, one can see a pattern, a method Moore could use to generate Open Question-type arguments in further cases.

4.3 The is-ought gap

4.3.1 Hume's law: an initial statement

Closely related to Moore's refutation of analytic reductionism is another doctrine in metaethics known as that of the 'is-ought gap', or 'Hume's Law'. This is the idea that it is impossible to validly derive an evaluative conclusion from wholly non-evaluative premises—or, as it is sometimes said, one cannot deduce conclusions about what ought to be from premises about what is. The locus classicus of this idea is the following well-worn passage from David Hume:

In every system of morality, which I have hitherto met with, I have always remark'd, that the author proceeds for some time in the ordinary way of reasoning, and establishes the being of a God, or makes observations concerning human affairs; when of a sudden I am surpriz'd to find, that instead of the usual copulations of propositions, is, and is not, I meet with no proposition that is not connected with an ought, or an ought not. This change is imperceptible; but is, however, of the last consequence. For as this ought, or ought not, expresses some new relation or affirmation, 'tis necessary that it shou'd be observ'd and explain'd; and at the same time that a reason should be given, for what seems altogether inconceivable, how this new relation can be a deduction from others, which are entirely different from it.¹⁴

Note that Hume is not merely observing that from 'x is the case' one cannot validly infer 'x ought to be the case'. Rather, he is making the general point that no collection of facts about what is the case entail any claim about what ought to be. Another way of stating the principle is to say that any (non-evaluative) description of the world can consistently be combined with any value system, or even with nihilism.¹⁵

Consider a simple challenge to Hume's Law. Suppose I pronounce that communism is bad. You ask me why I believe this. I say: 'It caused tens of millions of deaths, widespread poverty, and a great deal of human suffering. It is reasonable to believe that it would continue to have such effects in the future. Therefore, it is a bad system of government'. Assume that my historical claim is adequately established, and assume that you accept my generalization that communism can be expected to continue to have similar effects. Is not my conclusion thereby justified? Surely it is not merely some sort of simple logical fallacy I have committed. Yet the premises of my argument are entirely non-evaluative, while the conclusion is evaluative.

According to the standard doctrine, which I endorse, the argument fails to bridge the is-ought gap because it contains certain implicit premises which are evaluative. Specifically, the argument implicitly assumes that life is good and that poverty and human suffering are bad—if we do not assume these things, then the conclusion that communism is bad does not follow. Now, these assumptions may be perfectly reasonable, but they are, nevertheless, evaluative premises; therefore, the argument in question is not an example of a derivation of an evaluative conclusion from wholly non-evaluative premises.¹⁶ A similar point applies to all of the familiar ways in which we typically argue for evaluative conclusions; we generally argue that x is good by presupposing some evaluative standards. If we know that y is good, then we may argue that x is good by showing that x leads to y. If we know that people should do A when in circumstances of type C, then we may argue that S should do A by showing that S is in C.

This shows that some familiar moral arguments fail to bridge the is-ought gap. But what justifies the general conclusion that no argument can possibly do so? I think the initial justification for Hume's Law is intuitive: it simply is very difficult to see, on the face of it, how some mere description of how the world in fact is (in non-evaluative respects) could rule out some otherwise possible value system. It is equally difficult to see how a nihilist who recognized the natural properties of things but failed to value anything could be convicted of contradicting himself; it seems that nihilism would always remain a coherent position in the face of any purely natural facts. I do not say that it would be a plausible or justified position, but it would be coherent.

Another reason philosophers have given for the existence of the is-ought gap is that it is just a special case of a general principle of logic: one cannot validly deduce a conclusion that applies a predicate FP , from premises that nowhere contain that predicate.¹⁷ This is nothing special about evaluative predicates; it applies to any predicate whatsoever. For instance, one cannot infer a conclusion stating that something is red from premises that nowhere contain the word 'red'.

Before turning to objections to Hume's Law, it is important to understand what I take the doctrine's significance to be. I do not claim that Hume's Law refutes ethical reductionism.¹⁸ Rather, I see Hume's Law as an important lemma in my larger argument that one cannot account for moral knowledge without appealing to something like ethical intuition. To that end, I argue: (a) that we cannot know moral truths by observation, (b) that we cannot know (non-trivial) moral truths by deducing them from non-moral truths, (c) that we cannot know (non-trivial) moral truths by conceptual analysis, and (d) that we cannot know moral truths by scientific reasoning or inference to the best explanation. This understanding of the use I make of Hume's Law has implications for how we should treat alleged exceptions to the Law: such exceptions are important only if they represent possible ways of coming to know moral truths on the basis of non-moral truths. Thus, if there should prove to be a valid deductive inference containing only non-evaluative premises and an evaluative conclusion, but for some reason this inference could not be the source of our knowledge of the conclusion, then it would be irrelevant to my purposes here. It would be an exception to Hume's Law as traditionally stated, but not one that damages my argument in this chapter.

***4.3.2 Searle's challenge**

John Searle asks us to consider the following sequence of statements:

1. Jones uttered the words, 'I hereby promise to pay you, Smith, five dollars'.
2. Jones promised to pay Smith five dollars.
3. Jones placed himself under (undertook) an obligation to pay Smith five dollars.
4. Jones is under an obligation to pay Smith five dollars.
5. Jones ought to pay Smith five dollars.

(1) by itself does not entail (2); in order to conclude that Jones promised to pay Smith five dollars, we must have a body of background knowledge about the conventions in Jones' society (those that determine the meanings of Jones' words, how promises are made, etc.) and the circumstances of Jones' utterance (Jones was not joking, reading lines for a play, etc.). But this additional information consists entirely of natural facts. So let us just imagine that this information is all added to the premises of the argument, and ask whether the argument then would bridge the is-ought gap. (3) follows from (2), Searle says, because it is part of the concept of promising that a promise is the undertaking of an obligation; this is what distinguishes promising to do something from merely predicting that one will do it or expressing an intention to do it.

From (3) it follows at least that other things being equal (provided nothing else happens to remove the obligation), Jones is under an obligation to pay Smith five dollars.

From (4) it follows that, other things being equal (provided there is no more important, conflicting obligation), Jones ought to pay Smith five dollars. Thus, it seems that we can at least get some such conclusion as 'Other things being equal, Jones ought to pay Smith five dollars' from non-evaluative premises.^[19]

What is wrong with this argument? To begin with, let us distinguish two senses of 'undertook', which I will call the strong sense and the weak sense. In the strong sense, to undertake an obligation is to successfully place oneself under an obligation, such that it logically follows that one is in fact under an obligation. In the weak sense, to undertake an obligation is, roughly, to purport to place oneself under an obligation and/or to agree to accept such an obligation. (The exact

analysis is not important here.) A person who undertakes an obligation in the weak sense need not ever actually have an obligation. Consider an analogy: suppose you call me collect, and I agree to accept the charges, but due to a mistake, the phone company never actually charges me. My having 'accepted' the charges does not entail that I ever actually receive the charges. In a similar sense, a person might conceivably 'accept' an obligation without ever actually having the obligation. Why might this happen? Well, suppose, for example, that nihilism is true: that there are no moral properties whatsoever. Then although people frequently 'accept' and 'undertake' (in the weak sense) obligations, their undertakings would always fail to actually place them under obligations.

Of course, I believe that nihilism is false and that we do have obligations to keep our promises. But Searle has not refuted nihilism, and so he has not proven an 'ought' statement. There is no purely natural fact that the nihilist is unable to recognize. The nihilist, confronted with Searle's derivation, should grant that Jones made a promise, when this is understood as the acceptance or undertaking of an obligation in the weak sense; but he should deny that Jones undertook an obligation in the strong sense of successfully placing himself under an obligation.

Searle might argue that the weak notion of undertaking obligations does not do justice to our concept of 'promising'; if so, the nihilist should say, then no one ever really promises; at most we purport to promise. The conventions of our society may dictate that saying such words as 'I hereby promise to pay you five dollars' is taken as creating an obligation, but mere conventions cannot make it the case that such things as 'obligations' really exist; we cannot render nihilism false by mere fiat.

4.3.3 Geach's challenge

Peter Geach proposes the following alleged counter-example to Hume's Law:

1. If Evan were to promise to adopt some practice then he would adopt it. (Premise.)
2. If Evan were to utter sentence W, he would be promising to adopt the practice of doing something wrong twice a day. (Premise.)
3. Nobody should adopt the practice of doing something wrong twice a day. (Premise.)
4. If Evan were to utter W, he would adopt the practice of doing something wrong twice a day. (From 1, 2.)
5. Therefore, Evan should not utter W. (From 3, 4.)²¹

We imagine W to be some such sentence as, 'I promise henceforth to do two wrong things each day', and we imagine Evan to be a person with a very strong commitment to promise-keeping, such that (1) is true. The above appears to be a valid inference from premises (1), (2), and (3) to conclusion (5). (5) is clearly evaluative. (1) and (2) seem to be non-evaluative. If (3) were non-evaluative, the counter-example would seem to be in good order, but in fact (3) is evaluative. Nevertheless, Geach observes that (3) is an analytic truth. By definition, no one should do what is wrong; a fortiori, no one should do something wrong twice a day. So at least Geach seems to have shown how to go from non-evaluative statements and/or tautologies to a non-trivial evaluative conclusion.

Geach's example has at least two flaws.²² First, the argument is invalid. I assume that the inference from (3) and (4) to (5) is valid only if the following is a valid form of inference:

6. S ought not to do A.
7. If S did B, S would do A.
8. Therefore, S ought not to do B.

To see the problem with this, imagine the following scenario. Jon is a judge about to pass sentence on Mary, a convicted marijuana dealer. Mary's crime is minor at worst; John, however, has an intense, irrational hatred for all drug users, as a result of which he is determined to sentence Mary to either life imprisonment or death. He could sentence Mary to only a brief prison term, but he would not in fact do so. Now consider the following inference:

9. John ought not to sentence Mary to life imprisonment.
10. If John were to refrain from sentencing Mary to death, then he would sentence Mary to life imprisonment.

11. Therefore, John ought not to refrain from sentencing Mary to death (that is, he ought to sentence Mary to death).

(9) is true because what John ought to do is to sentence Mary to a brief prison term at most. (10) is true because of what we have stipulated about John's psychology. But (11) is false. In general, this form of inference is invalid because it is possible for there to be a situation in which what one should do is to do B and refrain from doing A, but what one actually would do after doing B is to do A as well. In such a situation, (6) and (7) are true and (8) is false.

The second problem with Geach's example is that the premise, 'If Evan were to promise to adopt some practice then he would adopt it', is evaluative. The relevant reading of (1) is:

$$\forall x (\text{Evan promises to do } x \rightarrow \text{Evan does } x)$$

('for all x , if Evan were to promise to do x , then Evan would do x '). This is evaluative because the following substitution instance of it is evaluative:

$$\text{Evan promises to act wrongly} \rightarrow \text{Evan acts wrongly.}$$

That is evaluative because, once we fix the natural facts about how Evan would behave upon making such a promise, whether the whole sentence is true then depends on whether that behavior would be wrong. For example, let's suppose that, if Evan were to promise to act wrongly, he would subsequently (thinking these things to be wrong) steal a piece of candy from his five-year-old niece and order a hamburger from McDonald's. Those are the natural facts about how Evan would behave. Whether ' $\text{Evan promises to act wrongly} \rightarrow \text{Evan acts wrongly}$ ' is true then depends on whether stealing the candy or ordering the hamburger would in fact be wrong. If they would, then the quoted sentence is true; if they would not (and no other thing Evan would do would be wrong), then the quoted sentence is false. That is why ' $\text{Evan promises to act wrongly} \rightarrow \text{Evan acts wrongly}$ ' is evaluative. This is an application of the criterion for evaluative statements developed below in section 4.3.5.

*4.3.4 Prior's challenge

A. N. Prior offers the following inference allegedly spanning the is-ought gap:

(A) 1. Tea-drinking is common in England.

2. Therefore, tea-drinking is common in England, or all New Zealanders ought to be shot.²³

The inference is perfectly valid, 'ought' appears in the conclusion, but it does not appear in the premise.

We might respond by denying that the conclusion is genuinely evaluative. But this would only open up the possibility of a different inference bridging the is-ought gap:

(A') 1. Tea-drinking is not common in England.

2. Tea-drinking is common in England, or all New Zealanders ought to be shot.

3. Therefore, all New Zealanders ought to be shot.

If (2) is evaluative, then argument (A) spans the is-ought gap; if (2) is non-evaluative, then (A') spans the is-ought gap.

However, this point has little bearing on the problem of moral knowledge, as neither of these inferences provides a plausible general model of moral knowledge. Although (A) may give us knowledge of an 'evaluative conclusion' in some sense, it is not an interesting evaluative conclusion. It is not one that, for example, we might use to guide our action, nor does it entail that anything actually has any evaluative property. The conclusion of (A'), on the other hand, is an interesting evaluation—but (A') provides no model for moral knowledge since there is no obvious way in which we could know both of its premises. If (1') were true, then we could readily discover it by empirical means, but there is no apparent way in which we could simultaneously discover (2) to be true without relying on an evaluative assumption.

What would be helpful would be an argument that combined the virtues of (A) and (A'): an argument that started from something obviously empirically verifiable, like (1), and concluded with an interesting, action-guiding evaluation, like (3). But Prior's example suggests no way of constructing such an argument. Obviously, we cannot simply combine the premises of (A) and (A'): that would indeed give us a valid argument from non-evaluative premises to an evaluative

conclusion. But since the premises would be inconsistent, this argument would provide no plausible model of how we come to know moral truths.

*4.3.5 Karmo's proof

The preceding response to Prior may leave one worried: once it has been shown that the is-ought barrier is not in principle impregnable, how can we be sure that every argument penetrating that barrier will prove as useless in accounting for moral knowledge as Prior's example? Can we devise a general proof for some useful form of Hume's Law?

Toomas Karmo has provided such an argument.²⁴ Assume that there is both a class of uncontroversially evaluative statements which clearly attribute evaluative properties to things—such as 'All New Zealanders ought to be shot' and 'Pleasure is good'—and a class of uncontroversially descriptive statements which clearly attribute only non-evaluative properties—such as 'Tea-drinking is common in England'. There are also some statements whose status is initially unclear, such as 'Tea-drinking is common in England, or all New Zealanders ought to be shot' and 'Everything Alfie says is true'. The latter statements, Karmo says, may be evaluative or non-evaluative, depending on what the world is like in non-evaluative respects. For instance, if tea-drinking is in fact common in England, then 'Tea-drinking is common in England, or all New Zealanders ought to be shot' is non-evaluative (roughly, because its truth does not then depend on whether New Zealanders ought to be shot); otherwise, the statement is evaluative. Similarly, if Alfie has in fact made some evaluative statements and has made no false descriptive statements, then 'Everything Alfie says is true' is evaluative, since its truth then depends on whether Alfie's evaluative statements were true; otherwise, it is non-evaluative.

Following is a more precise statement of Karmo's idea. Let a 'value system' be a maximal consistent conjunction of uncontroversially evaluative statements (that is, a conjunction of uncontroversially evaluative statements that is consistent and such that no further such statements could be added to it without creating an inconsistency). Similarly, let a 'possible world' be a maximal consistent conjunction of uncontroversially descriptive statements. The 'actual world' is the possible world that is true (the correct description of the world). I assume the following principle, which I shall call the Determinacy Assumption:

Determinacy A possible world together with a value system are consistent with at most one assignment of truth-values for all statements.

That is, if w is a possible world, v is a value system, and S is any statement, then either $(w \wedge v) \models S$, or $(w \wedge v) \models \neg S$.²⁵

Now, for statements whose status as evaluative or descriptive is initially unclear, classify them as follows: a statement S is evaluative in possible world w if, given w , the truth-value of S depends on which value system is correct. In other words, S is evaluative in w provided that there are two value systems such that one of them, combined with the uncontroversially descriptive statements in w , implies that S is true, while the other, combined with the same descriptive statements, implies that S is false. This characterization yields the result, for example, that 'Everything Alfie says is true' is evaluative in any possible world in which Alfie has made an evaluative statement and has made no false descriptive statements, but non-evaluative in any other possible world.

What we would like to see is whether there can be an argument such that

- i) Its premises are actually true,
- ii) Its premises are non-evaluative in the actual world,
- iii) Its premises entail its conclusion,
- iv) Its conclusion is evaluative in the actual world, and
- v) Its conclusion is non-trivial.

Conditions (ii), (iii), and (iv) are required for the argument to count as spanning the is-ought gap. Condition (i) is required for the argument to count as a way of gaining knowledge. Condition (v) is required for the argument to be interesting vis-à-vis the problem of moral knowledge—the conclusion should not be a mere tautology such as 'Right acts ought to be performed' or 'Bad things are bad' (these statements trivially follow from any set of premises). I assume that an evaluative statement is non-trivial only if there is at least one value system that rejects it.

Now assume for reductio that there is an argument satisfying conditions (i)–(v). Let $P_1, P_2, \dots$ be the premises of this argument, and let C be its conclusion. Let w be the actual world. Then we have the following:

1. $P_1, P_2, \dots$ entail C . (Condition (iii).)
2. C is non-trivial and evaluative in w . (Conditions (iv) and (v).)
3. There exist value systems v_1 and v_2 such that $(w \wedge v_1)$ entails C and $(w \wedge v_2)$ entails $\neg C$. (From 2.)
4. $\neg C$ entails that at least one of $P_1, P_2, \dots$ is false. (From 1.)
5. $(w \wedge v_2)$ entails that at least one of $P_1, P_2, \dots$ is false. (From 3, 4.)
6. There is a specific premise in $\{P_1, P_2, \dots\}$, call it P_i , such that $(w \wedge v_2)$ entails $\neg P_i$. (From 5 and Determinacy.)
7. w conjoined with the correct value system entails P_i . (From condition (i) and Determinacy.)
8. P_i is evaluative in w . (From 6, 7.)

Conclusion (8) follows from (6) and (7) because, given w , the truth of P_i turns on what the correct value system is — P_i is true according to the correct value system but would be false according to v_2 . But (8) contradicts condition (ii). Thus, there is no argument satisfying all of conditions (i)–(v).

I take it that Karmo's proof succeeds. However, Stephen Maitzen has objected to the argument.²⁶ Without going into detail, I note that Maitzen's objection turns on a criticism of Karmo's taxonomy, a criticism that depends on the assumption that nihilism—defined as the thesis that every evaluative statement is false—is logically possible. Among other things, Maitzen argues that Karmo's taxonomy is defective since it implies that the following statement is evaluative in every possible world:

B. Every evaluative statement is false, or torturing babies is wrong.

According to Karmo's taxonomy, this is evaluative in every possible world, since there is always a value system that makes the statement true (any value system that forbids baby-torture makes (B) true) and another value system that makes the statement false (any value system that endorses baby-torture will make both disjuncts of (B) false). But Maitzen says the statement should be classified as non-evaluative in at least some possible worlds, for:

1. Nihilism is logically possible. (Premise.)
2. Nihilism entails (B). (By V-introduction.)
3. Therefore, it is possible that nihilism and (B) both hold. (From 1, 2.)
4. In any possible world in which nihilism and (B) both hold, (B) is non-evaluative. (In such a world, every evaluative statement is false, since nihilism holds. But (B) also holds, so (B) must not be an evaluative statement.)
5. So there is at least one possible world in which (B) is non-evaluative. (From 3, 4.)

This example, Maitzen says, 'immediately falsifies Karmo's taxonomy of ethical sentences'.²⁷

Maitzen is mistaken because nihilism—understood as the thesis that every 'evaluative statement' in Karmo's sense is false—is logically impossible. One reason for this is that nihilism itself seems to be an evaluative statement;²⁸ therefore, in denying all evaluative statements, it contradicts itself. Another reason is that in any possible world, both 'Torturing babies is wrong' and 'It is not the case that torturing babies is wrong' are evaluative statements in Karmo's sense. But it is logically necessary that at least one of these two statements be true. So it is logically impossible that all evaluative statements be false. Of course, nihilism in some other sense might be logically possible—for example, perhaps it is possible that nothing has an evaluative property. But this would not refute Karmo's argument. Similarly, it might be that Karmo's sense of 'evaluative statement' differs from established usage—certainly it differs from Maitzen's usage—but this would not show Karmo's argument to be unsound either. In general, there is no question of 'falsifying' Karmo's

characterization of evaluative statements, as long as that characterization is understood—as I believe it should be—as a stipulative definition.

Given that the definition is purely stipulative, one may legitimately wonder whether Karmo's proof is relevant to what we were initially interested in—in particular, does Karmo's version of the isought gap help to establish that there is a problem in accounting for moral knowledge? I think it does. Karmo shows that there is no way of knowing evaluative statements by deduction from non-evaluative statements, in his sense of 'evaluative statements'. But Karmo's sense includes all the statements traditionally considered 'evaluative' by moral philosophers; thus, if it is difficult to know evaluative statements in Karmo's sense, it is also difficult to know evaluative statements in the traditional sense. Nor does Karmo appear to have expanded the sense of 'evaluative statement' so much that we might know some evaluative statements, for example, by observation: in Karmo's sense, an evaluative statement is one whose truth, once all the descriptive facts are fixed, still turns on which value system is correct. Barring appeals to such things as moral intuition, it remains difficult on the face of it to see how we might know such statements non-inferentially. Since Karmo has shown that we also cannot in general know such statements by deduction from non-evaluative statements, there is an interesting problem of moral knowledge.

In the next section, we consider attempts to show that we can know evaluative truths by non-deductive inferences from non-evaluative truths.